sMDD (k = ceil(N/2)) vs MDD — DD Python, paired rows (same instance, DD type, width and reduction)
MDD: results_python_01_11_2026_17-48-09.csv | sMDD: results_python_semi_semi_01_11_2026_17-48-09.csv
Columns are means over the paired rows; 'ratio' is the geometric mean of sMDD/MDD per row. Time = construction + reduction; Solve = longest/shortest path over the built diagram.
Bound means skip rows where either value is ±inf (empty restricted / unreachable terminal), counted in 'non-fin.'. 'sMDD +/=/-': rows where the sMDD bound is better / equal / worse.

Problem	DD	Red.	N	Time MDD	Time sMDD	ratio	Solve MDD	Solve sMDD	ratio	Nodes MDD	Nodes sMDD	ratio	Arcs MDD	Arcs sMDD	ratio	Bound MDD	Bound sMDD	non-fin.	sMDD +/=/-
Knapsack	Exact	no	21	7.55	17.39	1.78×	2.67	4.10	1.76×	906,750	877,435	0.54×	1.69e+06	5.54e+06	2.25×	7,599	7,599	0	0/21/0
	Exact	yes	21	9.49	24.52	2.07×	2.55	5.70	2.28×	872,545	872,545	1.00×	1.64e+06	5.51e+06	2.71×	7,599	7,599	0	0/21/0
	Restricted	no	155	12.39	21.41	1.66×	4.04	5.99	1.72×	1.69e+06	1.66e+06	0.61×	2.84e+06	5.2e+06	1.82×	14,671	14,671	0	0/155/0
	Restricted	yes	153	13.59	24.53	1.85×	3.56	5.74	1.95×	1.55e+06	1.55e+06	0.97×	2.59e+06	4.7e+06	2.06×	14,136	14,136	0	0/153/0
	RelaxPriority	no	156	15.49	21.46	1.48×	4.12	5.37	1.57×	1.81e+06	1.79e+06	0.62×	3.06e+06	4.67e+06	1.65×	591,205	590,478	0	55/101/0
	RelaxPriority	yes	156	18.27	26.84	1.63×	4.15	6.26	1.76×	1.77e+06	1.75e+06	0.95×	3e+06	4.59e+06	1.80×	591,205	590,478	0	55/101/0
Independent Set	Exact	no	12	0.4952	5.53	2.02×	0.1442	0.1504	0.88×	88,554	17,328	0.62×	112,731	363,090	1.16×	30.83	30.83	0	0/12/0
	Exact	yes	12	0.5884	5.94	1.87×	0.1440	0.1504	0.87×	88,554	17,328	0.62×	112,731	363,090	1.16×	30.83	30.83	0	0/12/0
	Restricted	no	248	2.85	43.25	7.37×	0.3834	0.4851	1.14×	284,321	232,699	0.80×	308,843	394,325	1.07×	28.17	23.57	0	0/103/145
	Restricted	yes	248	3.09	43.88	6.89×	0.3823	0.4799	1.09×	281,966	223,840	0.77×	306,030	366,651	1.00×	28.17	23.57	0	0/103/145
	RelaxPriority	no	301	9.18	39.79	2.73×	1.18	1.37	1.10×	843,945	870,929	0.95×	912,748	1.1e+06	1.14×	288	305	0	84/61/156
	RelaxPriority	yes	301	10.16	41.02	2.57×	1.12	0.4969	0.48×	694,886	278,086	0.44×	754,561	356,827	0.52×	288	305	0	84/61/156
Set Cover	Exact	no	178	0.3525	6.07	2.54×	0.0527	1.29	4.54×	39,335	34,222	0.99×	76,791	2.29e+06	6.16×	3.61	3.61	0	0/178/0
	Exact	yes	176	0.3563	5.90	3.46×	0.0427	0.4855	3.67×	32,494	31,116	1.00×	63,975	1.15e+06	5.97×	3.61	3.61	0	0/176/0
	Restricted	no	1930	8.35	14.48	2.57×	0.2691	1.08	3.39×	168,455	203,485	1.09×	326,064	1.77e+06	3.80×	8.76	8.76	0	2/1925/3
	Restricted	yes	1928	8.75	16.56	3.11×	0.2527	0.5536	2.52×	154,081	140,399	1.01×	300,757	981,498	3.12×	8.81	8.81	0	3/1922/3
	RelaxPriority	no	1902	7.30	18.06	3.02×	0.2245	0.9857	3.44×	145,933	185,181	1.14×	289,300	1.78e+06	4.09×	2.60	2.52	0	14/1736/152
	RelaxPriority	yes	1899	7.37	19.88	3.62×	0.2174	0.4918	2.45×	129,371	117,804	1.01×	257,385	948,006	3.21×	2.60	2.52	0	14/1733/152
SOC Knapsack	Exact	no	79	1.75	13.77	3.77×	0.3957	1.93	2.95×	195,787	6,296	0.08×	326,844	4.55e+06	6.08×	771	771	0	0/79/0
	Exact	yes	77	1.78	13.16	4.70×	0.0043	0.6453	25.93×	4,590	4,208	0.97×	8,847	1.73e+06	32.93×	762	762	0	0/77/0
	Restricted	no	526	0.2308	9.46	8.78×	0.0426	1.06	7.46×	31,501	6,350	0.24×	38,822	2.5e+06	14.12×	687	772	0	346/179/1
	Restricted	yes	520	0.2436	9.47	10.93×	0.0011	0.3472	36.62×	1,522	4,125	1.87×	2,604	926,881	46.53×	685	768	0	340/179/1
	RelaxPriority	no	529	1.14	9.62	4.45×	0.0533	0.7462	4.01×	34,826	8,156	0.24×	50,951	1.97e+06	6.55×	891	865	0	248/279/2
	RelaxPriority	yes	524	1.08	9.53	5.56×	0.0015	0.2706	12.89×	1,803	2,790	1.06×	3,274	747,583	14.99×	890	863	0	243/279/2
Sequencing	Exact	no	31	0.0262	0.1092	1.76×	0.0150	0.0085	0.61×	2,511	959	0.46×	11,234	5,347	0.57×	889	889	0	0/31/0
	Exact	yes	31	0.0326	0.1124	1.64×	0.0057	0.0055	0.98×	511	511	1.00×	2,517	2,517	1.00×	889	889	0	0/31/0
	Restricted	no	470	2.26	30.61	4.44×	0.2731	0.6017	1.30×	27,098	22,215	0.69×	140,090	124,705	0.80×	186,452	181,230	0	203/267/0
	Restricted	yes	469	2.38	30.35	4.15×	0.0949	0.4373	2.40×	5,811	12,704	1.62×	26,872	65,893	1.79×	186,845	181,611	0	203/266/0
	RelaxPriority	no	391	1.80	24.31	4.27×	0.6116	20.22	4.71×	17,090	12,763	0.63×	353,722	6.32e+06	3.56×	22,370	22,296	0	22/171/198
	RelaxPriority	yes	395	2.10	33.88	4.75×	0.1991	4.63	5.13×	2,613	7,251	1.93×	42,456	878,973	4.80×	22,221	22,179	0	23/196/176